

Lecture 03B

Linear Combination, Span, Linearly Independent, Basis

MH1201 Linear Algebra 2

Let $S = \{u, v\}$.

Which of the following vectors belong to $\text{span}(S)$?

Choose ALL correct answers

$$\text{span}(S) = \{ \lambda u + \mu v : \lambda, \mu \in \mathbb{R} \}$$

✓ 1. $u + v$ $\lambda=1, \mu=1$

✓ 2. $u - v$ $\lambda=1, \mu=-1$

✓ 3. 0 $\lambda=0, \mu=0$

✓ 4. u $\lambda=1, \mu=0$

✓ 5. $1048576v$ $\lambda=0, \mu=1048576$

Lecture 03B

- Linear Combination and Span $\leftarrow$ last week
- Linear Independent $\left\{ \begin{array}{l} \text{Today} \end{array} \right.$
- Basis

How to Describe a Vector Space?

- Start with a finite set of vectors and provide rules to combine them.

- How to combine these vectors?

— Pick some S and consider $\text{span}(S)$.

- How small can this set be?

— i.e. $S = \{v_1, v_2, \dots, x_{104576}\}$

$S^+ = \{v_1, v_2, \dots, \cancel{v_{1024}}, \dots\}$ st $\text{span}(S) = \text{span}(S^+)$

Linear Combination, Span, Linearly Independent

Let $S = \{v_1, v_2, \dots, v_n\}$.

- **(Linear Combination).** u is a linear combination of vectors in S if we can write

$$u = \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n \text{ for some } \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F}.$$

- **(Span).** The span of S is the set of vectors

$$\text{span}(S) = \{\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n : \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F}\}.$$

- **(Linearly Independent).** The set S is linearly independent if

(*) $\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n = \mathbf{0}$ implies $\lambda_1 = \lambda_2 = \dots = \lambda_n = 0$.

i.e. The only solution to (*) is $\lambda_1 = \lambda_2 = \dots = \lambda_n = 0$

If S is not linearly independent, then S is linearly dependent.

Example.

- Is $\{(2,3,1), (1, -1,2), (7,3,7)\}$ linearly independent? *Yes*
- Is $\{(2,3,1), (1, -1,2), (7,3,8)\}$ linearly independent? *No*

$$\lambda_1(2,3,1) + \lambda_2(1, -1,2) + \lambda_3(7,3,7) = (0,0,0)$$

This implies that

$$2\lambda_1 + \lambda_2 + 7\lambda_3 = 0$$

$$3\lambda_1 - \lambda_2 + 3\lambda_3 = 0$$

$$\lambda_1 + 2\lambda_2 + 7\lambda_3 = 0$$

Solving this system, the only solution is

$$\lambda_1 = \lambda_2 = \lambda_3 = 0$$

$$\lambda_1(2,3,1) + \lambda_2(1, -1,2) + \lambda_3(7,3,8) = (0,0,0)$$

This implies that

$$2\lambda_1 + \lambda_2 + 7\lambda_3 = 0$$

$$3\lambda_1 - \lambda_2 + 3\lambda_3 = 0$$

$$\lambda_1 + 2\lambda_2 + 8\lambda_3 = 0$$

→ not all zeros

There is a nontrivial solution

$$\lambda_1 = -2, \quad \lambda_2 = -3, \quad \lambda_3 = 1$$

Example.

- Is $\{(2,3,1), (1, -1,2), (7,3,7)\}$ linearly independent? **Yes**
- Is $\{(2,3,1), (1, -1,2), (7,3,8)\}$ linearly independent? **No**

$$\lambda_1(2,3,1) + \lambda_2(1, -1,2) + \lambda_3(7,3,8) = (0,0,0)$$

This implies that

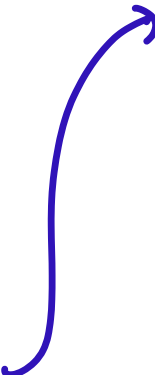
$$2\lambda_1 + \lambda_2 + 7\lambda_3 = 0$$

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$$\lambda_1 + 2\lambda_2 + 8\lambda_3 = 0$$

There is a nontrivial solution

$$\lambda_1 = -2, \quad \lambda_2 = -3, \quad \lambda_3 = 1$$


$$-2(2,3,1) - 3(1,-1,2) + (7,3,8) = (0,0,0)$$

$\Downarrow$

$$(7,3,8) = 2(2,3,1) + 3(1,-1,2)$$

Here, $(7,3,8)$ is a linear combination of $(2,3,1)$ and $(1,-1,2)$

Linear Dependence Lemma. Suppose $S = \{v_1, v_2, \dots, v_n\}$ is linearly dependent.

Then there exists $j \in \{1, 2, \dots, n\}$ such that

(a) $v_j \in \text{span}(\{v_1, v_2, \dots, v_{j-1}\})$. v_j is a linear combination of $v_1, \dots, v_{j-1}$

(b) Let $S^* = S \setminus \{v_j\}$. Then $\text{span}(S) = \text{span}(S^*)$.

↓ kick vector out ↪ span will still be the same

* This proof is wrong

a) Since S is linearly dependent, there is a nontrivial solution to $(*)$

$$(*) - \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n = 0$$

$$v_n = \frac{-\lambda_1 v_1}{\lambda_n} - \frac{-\lambda_2 v_2}{\lambda_n} - \dots - \frac{-\lambda_{n-1} v_{n-1}}{\lambda_n}$$

$$\therefore v_n \in \text{span}(v_1, v_2, \dots, v_{n-1})$$

wrong! cannot divide by λ_n because λ_n may be zero

Linear Dependence Lemma. Suppose $S = \{v_1, v_2, \dots, v_n\}$ is linearly dependent.

Then there exists $j \in \{1, 2, \dots, n\}$ such that v_j is a linear combi of $v_1, \dots, v_{j-1}$.

(a) $v_j \in \text{span}(\{v_1, v_2, \dots, v_{j-1}\})$.

(b) Let $S^* = S \setminus \{v_j\}$. Then $\text{span}(S) = \text{span}(S^*)$.

n) Proof : Since S is linearly dependent, there is a nontrivial solⁿ to (*)

Let j be the largest index s.t. $\lambda_j \neq 0$. $\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_j v_j = 0$

$$v_j = -\frac{\lambda_1}{\lambda_j} v_1 - \frac{\lambda_2}{\lambda_j} v_2 - \dots - \frac{\lambda_{j-1}}{\lambda_j} v_{j-1}$$

$$v_j \in \text{span}(v_1, \dots, v_{j-1})$$

Linear Dependence Lemma. Suppose $S = \{v_1, v_2, \dots, v_n\}$ is linearly dependent.

Then there exists $j \in \{1, 2, \dots, n\}$ such that

(a) $v_j \in \text{span}(\{v_1, v_2, \dots, v_{j-1}\})$.

(b) Let $S^* = S \setminus \{v_j\}$. Then $\text{span}(S) = \text{span}(S^*)$.

b) $\Rightarrow \text{span}(S^*) \subseteq \text{span}(S)$ because $S^* \subseteq S$
 $\Rightarrow \text{span}(S) \subseteq \text{span}(S^*)$

Let $u \in \text{span}(S) \Rightarrow u = \lambda_1 v_1 + \dots + \lambda_{j-1} v_{j-1} + \lambda_j v_j + \dots + \lambda_n v_n$.

$$= (\lambda_1 + \mu_j \lambda_j) v_1 + \dots + (\lambda_{j-1} + \mu_j \lambda_j) v_{j-1} + \lambda_{j+1} v_{j+1} + \dots + \lambda_n v_n$$

$$\Rightarrow u \in \text{span}(S^*)$$

Linear Dependence Lemma. Suppose $S = \{v_1, v_2, \dots, v_n\}$ is linearly dependent.

Then there exists $j \in \{1, 2, \dots, n\}$ such that

(a) $v_j \in \text{span}(\{v_1, v_2, \dots, v_{j-1}\})$.

(b) Let $S^* = S \setminus \{v_j\}$. Then $\text{span}(S) = \text{span}(S^*)$.

Linear Dependence Lemma 2.

Let $S = \{v_1, v_2, \dots, v_n\}$. Let $j \in \{1, 2, \dots, n\}$ and $S^* = S \setminus \{v_j\}$.

If $v_j \in \text{span}(S^*)$, then $\text{span}(S) = \text{span}(S^*)$.

↳ proof is the same as Linear Dependence Lemma (b).

Linear Dependence Lemma is just Gaussian Elimination

$$\begin{array}{ccc} \underline{v}_1 & \underline{v}_2 & \underline{v}_3 \\ \begin{bmatrix} 2 & 1 & 7 \\ 3 & -1 & 3 \\ 1 & 2 & 8 \end{bmatrix} & \xrightarrow{\text{r.r.e.f.}} & \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \end{array}$$

columns with leading entry column with no leading entry

- if rref is not identity, then $\{\underline{v}_1, \underline{v}_2, \underline{v}_3\}$ is linearly dependent
- columns with no leading entry belongs to the span of columns with leading entries.
- $\underline{v}_3 \in \text{span}(\underline{v}_1, \underline{v}_2)$

Tutorial 3, Q2. Prove the following statements.

- (a) $\{v\}$ is linearly independent if and only if v is a nonzero vector.
- (b) $\{u, v\}$ is linearly independent if and only if u is not a scalar multiple of v and v is not a scalar multiple of u .
- (c) If $0 \in S$, then S is linearly dependent.

- Go thru next Monday

Example. $\{1, x, x^2, \dots, x^d\}$ is linearly independent.

$$\lambda_0 \cdot 1 + \lambda_1 x + \lambda_2 x^2 + \dots + \lambda_d x^d = \underset{\sim}{0} \rightarrow \text{RHS. not zero, is the zero vector}$$

$$\Rightarrow \lambda_0 = \lambda_1 = \dots = \lambda_d = 0$$

$\therefore \{1, x, x^2, \dots, x^d\}$ is linearly independent

$\therefore$ a polynomial
whose coeff. are
all zero.

Basis

$$[a, b, c] \neq [b, a, c]$$

$$\{a, b, c\} = \{b, a, c\}$$

(Basis) Let $\mathcal{V}$ be a vector space.

The ordered list $\mathcal{B} \subseteq \mathcal{V}$ is a basis for $\mathcal{V}$ if

- $\text{span}(\mathcal{B}) = \mathcal{V}$, and
- the set $\mathcal{B}$ is linearly independent.

Recall: Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$.

- **(Span).** The span of S is the set of vectors

$$\text{span}(S) = \{\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n : \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F}\}.$$

- **(Linear Independent).** The set S is linearly independent if

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n = \mathbf{0} \text{ implies } \lambda_1 = \lambda_2 = \dots = \lambda_n = 0.$$

Example. Is $B = [1, x, x^2, \dots, x^d]$ a basis for $\mathcal{P}_d(\mathbb{R})$?

* $\text{span}(B) = \mathcal{P}_d(\mathbb{R})$?

$$\mathcal{P}_d(\mathbb{R}) \subseteq \text{span}(S)$$

$$\text{let } f(x) = a_0 + a_1x + a_2x^2 + \dots + a_dx^d \in \mathcal{P}_d(\mathbb{R})$$

$$\lambda_0 = a_0, \lambda_1 = a_1, \dots, \lambda_d = a_d$$

$$\Rightarrow f(x) \in \text{span}(S) \Rightarrow \mathcal{P}_d(\mathbb{R}) \subseteq \text{span}(S)$$

$$\therefore \mathcal{P}_d(\mathbb{R}) = \text{span}(S)$$

$$\lambda_0 \cdot 1 + \lambda_1 x + \dots + \lambda_d x^d = f(x)$$

* Is B linearly independent?

Yes! We showed it earlier.

$\therefore B$ is a basis for $\mathcal{P}_d(\mathbb{R})$

Example. Is $C = \left[1 - x^2, \frac{1}{2}(-x + x^2), \frac{1}{2}(x + x^2) \right]$ a basis for $\mathcal{P}_2(\mathbb{R})$?

* $\text{span}(C) = \mathcal{P}_2(\mathbb{R})$? Yes. We showed this last week

* Linearly Independent? Let $\lambda_0(1-x^2) + \lambda_1\left(-\frac{x}{2} + \frac{x^2}{2}\right) + \lambda_2\left(\frac{x}{2} + \frac{x^2}{2}\right) = 0$

$$\Rightarrow \left. \begin{array}{ll} \lambda_0 & = 0 \\ -\frac{\lambda_1}{2} + \frac{\lambda_2}{2} & = 0 \\ -\lambda_0 + \frac{\lambda_1}{2} + \frac{\lambda_2}{2} & = 0 \end{array} \right\} \begin{array}{l} \text{coef. } x^0 \\ \text{coef. } x^1 \\ \text{coef. } x^2 \end{array} \left. \begin{array}{l} \lambda_0 = \lambda_1 = \lambda_2 = 0 \\ \text{is the only solution.} \\ \therefore C \text{ is L.I.} \\ \therefore C \text{ is also a basis} \end{array} \right.$$

! $\mathcal{P}_2(\mathbb{R})$ has at least two diff bases!

$$B = [1, x, x^2]$$

$$C = \left[1 - x^2, \frac{1}{2}(-x + x^2), \frac{1}{2}(x + x^2) \right]$$

How to Describe a Vector Space $\mathcal{V}$?

- Start with a **finite set of vectors** S and hope that $\text{span}(S) = \mathcal{V}$.
- How small can the spanning set S be?

Ans: a smallest spanning set has d vectors where d is the number of vectors in a basis for $\mathcal{V}$.

We call d , the dimension of $\mathcal{V}$.

Dimension

Something is “wrong”
with this definition...



(Basis) Let $\mathcal{V}$ be a vector space.

The set $S \subseteq \mathcal{V}$ is a basis for $\mathcal{V}$ if

- $\text{span}(S) = \mathcal{V}$, and
- the set S is linearly independent.

The dimension of $\mathcal{V}$, denoted by $\dim(\mathcal{V})$, is given by the cardinality of a basis for $\mathcal{V}$.



For this definition to be “well-defined”,
we need different bases to have the same size

Proposition. Let $\mathcal{V}$ be a vector space.

Let S and I be finite subsets of $\mathcal{V}$.

If $\text{span}(S) = \mathcal{V}$ and I is linearly independent, then

$$|I| \leq |S|.$$

$$S = \{x_1, \dots, x_s\} \Rightarrow \text{span}(S) = \mathcal{V}$$

$$I = \{x_1, \dots, x_t\} \Rightarrow I \text{ is linearly independent}$$

Prop states $s \geq t$

– Proof via Gaussian
Elimination

– Colab has a proper proof

Gaussian Elimination. Let $\mathcal{V}$ be a vector space.

INPUT:

- $S = \{v_1, v_2, \dots, v_s\}$ such that $\text{span}(S) = \mathcal{V}$
- $I = \{u_1, u_2, \dots, u_t\}$ is linearly independent

OUTPUT: Ordered list of size s

$\mathcal{B} = [u_1, u_2, \dots, u_t, v_*, v_*, \dots, v_*]$
such that $\text{span}(\mathcal{B}) = \mathcal{V}$.

linearly dependent

$u_1, u_2, v_1, \dots, \cancel{v_i}, \dots, v_s$

$\Rightarrow$ This implies $s \geq t$

Example. Let $\mathcal{V} = \mathbb{R}^4$.

$$\mathbf{u}_1 = (1,0,1,1), \mathbf{u}_2 = (1,0,-1,1), \mathbf{u}_3 = (1,0,2,4)$$

$$\mathbf{v}_1 = (1,0,0,0), \mathbf{v}_2 = (0,1,0,0), \mathbf{v}_3 = (0,0,1,0), \mathbf{v}_4 = (0,0,0,1)$$

Then $I = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is linearly independent and

$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ spans $\mathbb{R}^4$.

Then Gaussian Elimination gives the list

$$\mathcal{B} = [\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{v}_2].$$

on colab it is shown that
this procedure works for
abstract values

When $\mathcal{V} = \mathbb{R}^m$, we do Gaussian Elimination aka row reduction.

$$\begin{array}{c} \text{I} \qquad \qquad \text{S} \\ \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & -1 & 2 \\ 1 & 1 & 4 \end{bmatrix}}_{\substack{\triangle \\ \triangle \\ \triangle}} \quad \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\substack{\triangle \\ \triangle}} \xRightarrow{\text{r.r.e.f.}} \begin{bmatrix} \triangle & \triangle & \triangle & & \triangle \\ 1 & 0 & 0 & 1 & 0 & 1/2 & 1/2 \\ 0 & 1 & 0 & 1/3 & 0 & -1/2 & 1/6 \\ 0 & 0 & 1 & -1/3 & 0 & 1 & 1/3 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

leading entry (pointing to the first row of the second matrix)

Theorem (All bases have the same size). Let $\mathcal{V}$ be a finite-dimensional vector space.

Let $\mathcal{B}$ and $\mathcal{C}$ be bases for $\mathcal{V}$. Then $|\mathcal{B}| = |\mathcal{C}|$.

Proof

We will show:

• $|\mathcal{B}| \geq |\mathcal{C}|$

• $|\mathcal{C}| \geq |\mathcal{B}|$ (DIY)

$\left. \begin{array}{l} \mathcal{B} \text{ is a basis for } \mathcal{V} \Rightarrow \text{span}(\mathcal{B}) = \mathcal{V} \\ \mathcal{C} \text{ is a basis for } \mathcal{V} \Rightarrow \mathcal{C} \text{ is linearly independent} \end{array} \right\} \Rightarrow |\mathcal{B}| \geq |\mathcal{C}|$

~~⊗~~ Be careful! Here we have made
some assumption.
We have assumed a basis exists

Finite-Dimensional Spaces

Let $\mathcal{V}$ be a vector space.

$\mathcal{V}$ is a finite-dimensional space if there is a finite set S such that S spans $\mathcal{V}$.

Examples of vector spaces that are NOT finite-dimensional

- Space of all polynomials
- Space of all functions $\mathcal{F}(X, \mathbb{R})$

(Basis) Let $\mathcal{V}$ be a vector space.

The set $S \subseteq \mathcal{V}$ is a basis for $\mathcal{V}$ if

- $\text{span}(S) = \mathcal{V}$, and
- the set S is linearly independent.

Theorem. Let $\dim(\mathcal{V}) = d$. ✖

Consider $S \subseteq \mathcal{V}$.

If $|S| = d$ and $\text{span}(S) = \mathcal{V}$,
then S is a basis for $\mathcal{V}$.

Theorem. Let $\dim(\mathcal{V}) = d$. ✖✖

Consider $S \subseteq \mathcal{V}$.

If $|S| = d$ and S is linearly independent,
then S is a basis for $\mathcal{V}$.

Theorem (Criteria for a Basis).

Let $\mathcal{V}$ be a vector space with $\dim(\mathcal{V}) = d$. Then
a set $S \subseteq \mathcal{V}$ is a basis for $\mathcal{V}$ if **two** out of the
following three holds.

(A) $|S| = d$,

(B) $\text{span}(S) = \mathcal{V}$,

(C) the set S is linearly independent.

If S is a basis, all three conditions hold.

In other words, if you know $\dim(\mathcal{V}) = d$,
you just need to check $|S| = d$.

AND [either (B) OR (C)].

Basis Reduction Lemma.

Let $\mathcal{V}$ be a finite-dimensional space.

Let S be a finite subset of $\mathcal{V}$.

If $\text{span}(S) = \mathcal{V}$, then there exists a subset $\mathcal{B} \subseteq S$ such that $\mathcal{B}$ is a basis of $\mathcal{V}$.

$S = \{v_1, \dots, v_s\}$ spans $\mathcal{V}$.

↓ Remove v_i 's

$\mathcal{B}$ is a basis for $\mathcal{V}$

Basis Extension Lemma.

Let $\mathcal{V}$ be a finite-dimensional space.

Let I be a finite subset of $\mathcal{V}$.

If I is linearly independent, then there exists a superset $\mathcal{B} \supseteq I$ such that $\mathcal{B}$ is a basis of $\mathcal{V}$.

$I = \{u_1, \dots, u_t\}$ linearly independent

↓ Insert v_i 's

$\mathcal{B} = \{u_1, \dots, u_t, v_1, \dots, v_s\}$
 $\mathcal{B}$ is a basis for $\mathcal{V}$.

Basis Reduction Lemma.

Let $\mathcal{V}$ be a finite-dimensional space.

Let S be a finite subset of $\mathcal{V}$.

If $\text{span}(S) = \mathcal{V}$, then there exists a subset $\mathcal{B} \subseteq S$ such that $\mathcal{B}$ is a basis of $\mathcal{V}$.

Proof Sketch.

Gaussian Elimination with $I \leftarrow \emptyset$ and $S \leftarrow S$

Basis Extension Lemma.

Let $\mathcal{V}$ be a finite-dimensional space.

Let I be a finite subset of $\mathcal{V}$.

If I is linearly independent, then there exists a superset $\mathcal{B} \supseteq I$ such that $\mathcal{B}$ is a basis of $\mathcal{V}$.

Proof Sketch.

Gaussian Elimination with $I \leftarrow I$ and $S \leftarrow \mathcal{B}^*$,
where $\mathcal{B}^*$ is some basis for $\mathcal{V}$.

Summary

Linear Dependence Lemma.

Suppose $S = \{v_1, v_2, \dots, v_n\}$ is linearly dependent.

Then there exists $j \in \{1, 2, \dots, n\}$ such that

- (a) $v_j \in \text{span}(\{v_1, v_2, \dots, v_{j-1}\})$.
- (b) Let $S^* = S \setminus \{v_j\}$. Then $\text{span}(S) = \text{span}(S^*)$.

Proposition. Let $\mathcal{V}$ be a vector space.

Let S and I be finite subsets of $\mathcal{V}$.

If $\text{span}(S) = \mathcal{V}$ and I is linearly independent, then

$$|I| \leq |S|.$$

Basis Reduction Lemma.

Let $\mathcal{V}$ be a finite-dimensional space.

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If $\text{span}(S) = \mathcal{V}$, then there exists a subset $\mathcal{B} \subseteq S$ such that $\mathcal{B}$ is a basis for $\mathcal{V}$.

Basis Extension Lemma.

Let $\mathcal{V}$ be a finite-dimensional space.

Let I be a finite subset of $\mathcal{V}$.

If I is linearly independent, then there exists a superset $\mathcal{B} \supseteq I$ such that $\mathcal{B}$ is a basis of $\mathcal{V}$.

Theorem (All bases have the same size).

Let $\mathcal{V}$ be a **finite-dimensional** vector space.

Let $\mathcal{B}$ and $\mathcal{C}$ be bases for $\mathcal{V}$. Then $|\mathcal{B}| = |\mathcal{C}|$.

Theorem.

Let $\mathcal{V}$ be a finite-dimensional space.

Then $\mathcal{V}$ has a basis of finite size.

The dimension of $\mathcal{V}$, denoted by $\dim(\mathcal{V})$, is given by the cardinality of a basis for $\mathcal{V}$.

Standard Bases

$$\mathbb{R}^3 \quad \begin{aligned} e_1 &= [1 \ 0 \ 0] \\ e_2 &= [0 \ 1 \ 0] \\ e_3 &= [0 \ 0 \ 1] \end{aligned}$$

- $[e_i : 1 \leq i \leq n]$ is the standard basis for $\mathbb{R}^n$. Here, e_i is the tuple with one at i -th position and zero elsewhere.

- $[E_{ij} : 1 \leq i \leq m, 1 \leq j \leq n]$ is the standard basis for $\mathcal{M}_{m \times n}(\mathbb{R})$. Here, E_{ij} is the matrix with one at (i, j) -th position and zero elsewhere.

↳ what is $\dim(\mathcal{M}_{m \times n})$?

- $[1, x, \dots, x^d]$ is the standard basis for $\mathcal{P}_d(\mathbb{R})$.

↳ what is $\dim(\mathcal{P}_d)$?